

NaijaSense AI Solution Paper

DSN × Bluechip Tech LLM Agent Challenge

Team TAOTECH SOLUTIONS

Abstract

NaijaSense AI is a dual-task LLM system: **Task A** simulates a domain-aware star rating and aligned first-person review from persona + product text; **Task B** ranks recommendations from a persona-only narrative with mandatory **Reason-Before-Recommend** agent_reasoning. Production agents are two POST endpoints (Groq router llama-3.1-8b-instant + generator llama-3.3-70b-versatile), grounded in a 311-row Yelp/Goodreads/Amazon corpus with Nigerian cold-start defaults. A unified hub (/unified) provides demos and ablations; evaluation uses the dedicated task URLs below.

Endpoints:

- Task A — <https://naija-sense-ai.vercel.app/task-a/user-modeling>
 - Task B — <https://naija-sense-ai.vercel.app/task-b/recommendation>
- Code — <https://github.com/taotechs/NaijaSense-AI>

1. Problem & Submission Contract

Reviews encode tone, rating bias, and domain taste. We target auditable agents—not opaque chat—with strict I/O per task.

Task	Path	Input	Output
A	POST /task-a/user-modeling	user_persona, product_details (strings)	rating, review_reasoning, review_text
B	POST /task-b/recommendation	{ user_id, persona } only	recommendations[], agent_reasoning

2. Architecture

flowchart TB C[Client] --> TA[Task A: two-pass agent] C --> TB[Task B: retrieve then rerank] TA --> R1[rating + reasoning + review] TB --> S1[Top-30 corpus pool] S1 --> S2[LLM rerank + agent_reasoning] S2 --> R2[ranked recommendations]

Task A (TaskATwoPassAgent). Parse texts → infer product domain (food/tech/etc.) from product details only → **Pass 1 (router):** JSON rating + rationale with domain few-shots → **Pass 2 (generator):** 2–4 sentence review with **rating locked**. Heuristic fallback if JSON fails.

Task B (TaskBPipelineAgent). Parse persona (location, budget, interests; no separate query) → **Stage 1:** top-30 candidates from catalog + Nigerian cold-start/cross-domain priors → **Stage 2 (router):** Reason-Before-Recommend monologue + JSON rankings (item_id, confidence_score); stage-1 fallback on parse error.

Demo hub (not scored). /unified adds intent routing, silent history by user_id, critique→regenerate, streaming NDJSON, and pidgin/Yoruba modes—used for UX and scripts/run_real_benchmark.py ablations only.

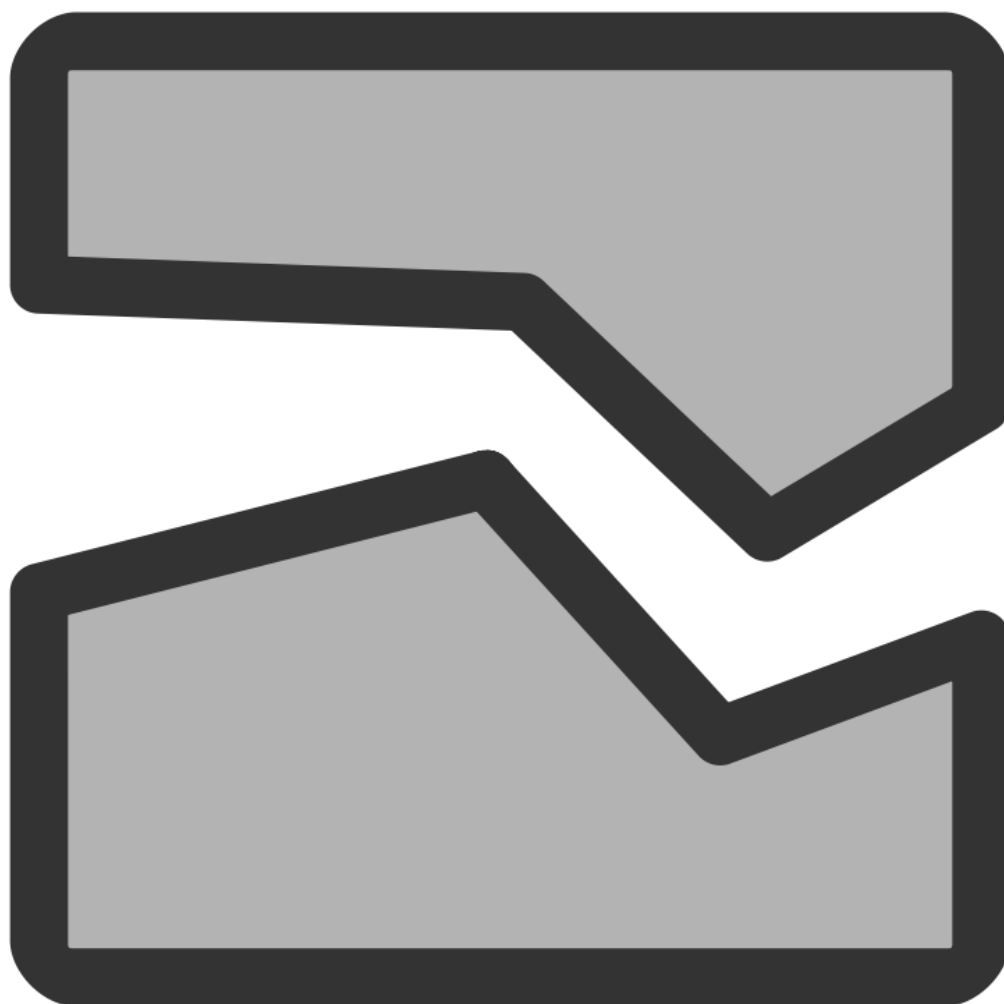


Figure 1. Behavioral Intelligence Hub output (supplementary demo at /unified).

3. Data, Models & Evaluation

Corpus: data/processed/review_corpus.jsonl — 311 rows (Yelp 274, Goodreads 31, Amazon 6). **Models:** Groq router + generator; ORCHESTRATOR_PROVIDER=groq. **Metrics:** Task A — ROUGE-1/L, token-F1 (BERTScore fallback), RMSE; Task B — NDCG@10, Hit@10 (evals.py, scripts/run_real_benchmark.py).

Task A ablations (orchestrator path, N=20): full ROUGE-1 0.161, RMSE 1.25; no_llm ROUGE-1 0.126 — LLM is the main lever. Submission adds **two-pass rating lock** to cut score-text drift. **Task B ablations** (legacy deterministic ranker, N=25): Hit@10 0.20 vs ~0.50 random on hard same-domain distractors; **submission** uses LLM rerank over stage-1 pool to address this.

4. Reproducibility & Limitations

```
git clone https://github.com/taotechs/NaijaSense-AI.git && cd NaijaSense-AI cp .env.example .env #
GROQ_API_KEY docker compose up --build python scripts/smoke_api.py http://127.0.0.1:8000
```

Swagger: /docs · Smoke tests cover both task endpoints. **Limits:** (1) Ablations ≠ submission pipelines—score /task-a and /task-b directly; (2) Task A persona must be self-contained (no silent history on submission route); (3) Task B rerank depends on stage-1 recall; (4) Amazon slice undersampled when HF was unavailable.

5. Conclusion

NaijaSense AI ships two deployable, explainable agents from one repo: Task A with domain-aware two-pass review simulation, Task B with corpus retrieval and mandatory agent_reasoning. One solution paper covers both tasks; each task has its own agent URL. The hub supports iteration; the endpoints in §1 are the primary evaluation surface.